

System and Software User Manual

STAR: Spatial Topography Accessibility Robot

Operator's Guide for Daily Use, Setup, and Maintenance

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Texas A&M University
Department of Electronic Systems Engineering Technology (ESET)
ESET Senior Capstone, Spring 2026

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April 28, 2026

Version 1.0 – Capstone Submission

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1 Scope

1.1 Purpose

This manual is the operator-facing companion to the STAR Software Description Document (SDD). The SDD explains HOW the STAR software is designed and built; this manual explains HOW TO USE it. If a procedure here is insufficient, consult the SDD for the underlying design, or consult the source repository for authoritative detail.

1.2 Audience

The intended readers are:

- Capstone reviewers validating that STAR can be operated by a non-developer through documented procedures.
- Follow-on student teams inheriting the robot and needing to bring it up on new hardware or at a new site.
- ADA facility auditors or building managers piloting STAR for a real compliance run.

Knowledge of Linux shells, SSH, and browser-based dashboards is assumed; knowledge of ROS2 internals, C firmware, or protobuf is not.

1.3 Out of Scope

This manual does not cover software architecture, algorithm derivations, data formats, testing strategy, or security design decisions. Those live in the STAR SDD. Mechanical assembly, PCB schematics, and BOM are covered in their own capstone deliverables.

1.4 Status Label Legend

Throughout this document, three status labels identify how complete each subsystem or feature is on the delivered system:

- **[IMPLEMENTED]** – complete and exercised in daily use.
- **[STRETCH]** – partial implementation with scaffolding; fit for demonstration but not yet complete.
- **[ARCHITECTED]** – designed and specified but not coded.

2 Referenced Documents

- STAR Software Description Document (companion capstone deliverable, `final_docs/software_description_`
- `docs/ARCHITECTURE.md` – end-to-end operations stack topology (Pi5, Tailscale, k3s, Cloudflare).
- `infra/README.md` – Pi5-side system files, Caddy configuration, systemd units, and fresh-Pi5 install procedure.
- `docs/PI_DEPLOYMENT.md` – ROS2 Jazzy workstation and Pi5 provisioning.
- `scripts/flash-rx72n.sh` – RX72N firmware flashing helper.

- `monitoring/grafana-star-dashboard.json` – Grafana dashboard definition used by the operator cockpit.
- `star-ros2/src/star_simple_bridge/` – MVP ROS2 bridge that serves Prometheus metrics and map images.
- `star-ros2/src/star_bringup/foxfordglove/star-default.json` – default Foxglove / Lichtblick layout.
- ADA 2010 Standards, Section 405.2 (Ramp Slope).

3 Software Summary

3.1 What STAR Does

STAR is an autonomous indoor mobile robot that performs ADA Title III compliance audits. The operator drives or dispatches the robot into a building; the robot maps the space, measures geometric features (ramp slope, path width, trip hazards), and produces a structured compliance report. All of the control software runs on a Raspberry Pi 5 on the robot; a second microcontroller (Renesas RX72N) performs real-time motor control. The operator interacts with STAR through web-based dashboards – primarily Grafana for control and Lichtblick (or Foxglove) for 3D visualization.

3.2 Operator-Visible Software Inventory

Table 1 summarizes what the operator actually sees and uses. Everything underneath is described in the SDD.

Table 1: Operator-visible software inventory.

Component	Where	What it does for the operator
Grafana dashboard	k3s cluster / browser	Primary cockpit. Live telemetry, autonomy toggle, e-stop, speed, SLAM controls. [IMPLEMENTED]
Lichtblick web app	k3s cluster / browser	3D visualization of LiDAR, map, odometry, and TF tree. Foxglove-compatible. [IMPLEMENTED]
Foxglove Studio desktop	Laptop (tailnet)	Power-user alternative to Lichtblick, same layout file. [IMPLEMENTED]
<code>star_cockpit_api</code>	Pi5 :9102	HTTP API backing the Grafana cockpit toggles. [IMPLEMENTED]
<code>star_simple_bridge</code>	Pi5 :9101	ROS2 serial bridge to RX72N, Prometheus metrics, map image server. [IMPLEMENTED]
Caddy (Pi5-side)	Pi5 :443	Local-CA TLS front for <code>star.local</code> . [IMPLEMENTED]
<code>star-slam-mvp.service</code>	Pi5 systemd	Autostarts the SLAM MVP node set on boot; safe MANUAL state. [IMPLEMENTED]
RX72N firmware	On-board MCU	Closed-loop motor control, encoder and IMU decoding. [IMPLEMENTED]
<code>flash-rx72n.sh</code>	Workstation	One-shot firmware flashing helper. [IMPLEMENTED]

3.3 Operator Connection Topology

Figure 1 summarizes how an operator in a browser reaches the robot. The Pi5 is on a Tailscale mesh (IP 100.64.0.6); a k3s homelab cluster hosts Grafana and Lichtblick; a public Caddy instance on the homelab fronts both for Cloudflare TLS.

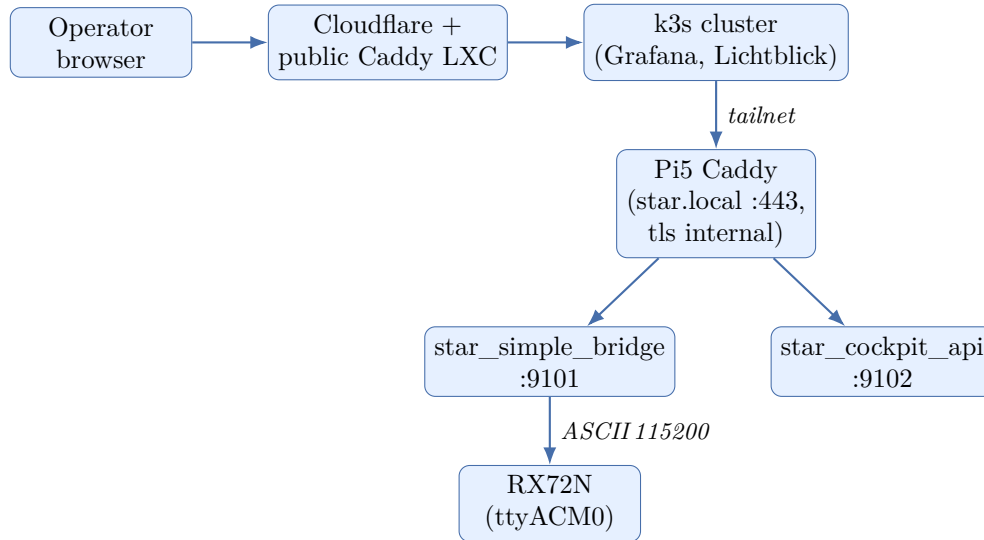


Figure 1: Operator connection topology. The Pi5 appears to the operator as `star.local`; Grafana and Lichtblick live in a homelab k3s cluster that is also on the Tailscale mesh.

4 Installation and First-Time Setup

This section covers procedures you perform once per site or robot. For daily operation see Section 5.

4.1 Prerequisites

- STAR robot with charged battery pack, RPLiDAR C1, BNO055 IMU, and four DRV8263H-driven motors.
- Raspberry Pi 5 running Ubuntu 24.04 with ROS2 Jazzy installed per `docs/PI_DEPLOYMENT.md`.
- Operator laptop with a modern browser and a Tailscale client enrolled in the same tailnet as the Pi5.
- USB-C power for the workstation, Renesas E2 Lite debugger for initial firmware flashing, and a USB-A to micro-USB cable for SCI boot-mode fallback.

4.2 Provisioning the Pi5

Follow `docs/PI_DEPLOYMENT.md` end-to-end. At a minimum:

1. Install ROS2 Jazzy from the OSRF apt repository and source `/opt/ros/jazzy/setup.bash`.
2. Install the STAR udev rules for `/dev/star-mcu` and `/dev/star-lidar` so device paths are stable across reboots.
3. Enroll the Pi5 in Tailscale and confirm its address resolves to 100.64.0.6 (or whichever tailnet address is documented for your team).

4. Build the ROS2 workspace with `./build-ros2.sh`.

4.3 Installing Pi5 Infrastructure Files

Refer to `infra/README.md`. Copy the system files into place on a fresh Pi5:

```
# From the repository root on the Pi5
sudo cp infra/pi5/opt/star_cockpit_api/server.py \
    /opt/star_cockpit_api/
sudo chmod +x /opt/star_cockpit_api/server.py

sudo cp infra/pi5/etc/caddy/Caddyfile /etc/caddy/Caddyfile
sudo cp infra/pi5/etc/systemd/system/*.service \
    /etc/systemd/system/

# Install Caddy (arm64)
sudo curl -fsSL -o /usr/local/bin/caddy \
    "https://caddyserver.com/api/download?os=linux&arch=arm64"
sudo chmod +x /usr/local/bin/caddy

sudo systemctl daemon-reload
sudo systemctl enable --now caddy.service
sudo systemctl enable --now star-cockpit-api.service
sudo systemctl enable --now star-slam-mvp.service
```

After this, rebooting the Pi5 brings up the complete operator stack automatically: Caddy on port 443, cockpit API on 9102, `star_simple_bridge` on 9101, and the SLAM MVP ROS2 node set in MANUAL mode with the emergency stop cleared.

4.4 Flashing the RX72N Firmware

Two flashing methods are supported by `scripts/flash-rx72n.sh`:

E2 Lite (default, recommended for lab work):

- SW4 Pin1 = OFF, Pin2 = OFF (Single Chip Mode).
- Board powered externally.
- FINE interface at 250 kbit/s.
- 1 Mbit/s FINE causes framing errors on the STAR PCB and must not be used.

SCI Boot Mode (field fallback via on-board USB-UART):

- SW4 Pin1 = ON, Pin2 = OFF.
- Power-cycle the board AFTER setting the switches.
- After a successful flash, return SW4 Pin1 = OFF and power-cycle again to run the new image.

```
# Build a release image and flash via E2 Lite
make build-rx72n-release
./scripts/flash-rx72n.sh \
    star-rx72n-firmware/build/release/star-rx72n.mot

# Fallback over on-board USB-UART (SCI Boot Mode)
```

```
./scripts/flash-rx72n.sh \
    star-rx72n-firmware/build/release/star-rx72n.mot \
    sci /dev/ttyACM0
```

4.5 Grafana and Lichtblick Deployment

Grafana, Prometheus, and Lichtblick are deployed as k3s workloads on the homelab cluster (Tailscale address 100.64.0.1). The Grafana dashboard is pushed into running Grafana via its HTTP API, so the authoritative source of truth is the JSON file at `monitoring/grafana-star-dashboard.json`. After any edit, re-pushing the dashboard is a one-step operation the cluster admin performs out of band.

The Grafana dashboard iframes the STAR map served by `star_simple_bridge` at `https://star.local/map.png`, which requires the operator's browser to trust the Pi5 Caddy's local CA. See `docs/ARCHITECTURE.md` for the one-time browser trust-store step.

4.6 Verifying the Install

From the operator's browser on the tailnet, confirm each surface is reachable:

- `https://star.local/` – Pi5 Caddy serves `star_simple_bridge`; `/map.png` returns an image (or a 404 until SLAM has produced a map).
- `https://star.local/api/state` – cockpit API returns `{autonomy, estop, speed, slam}` JSON.
- Grafana URL (deployed through public Caddy + Cloudflare) – the STAR dashboard loads and panels populate within seconds.
- Lichtblick URL (deployed through public Caddy + Cloudflare) – the default layout loads; click "Open Connection" and enter `ws://100.64.0.6:8765` to attach to the foxglove bridge on the robot.

5 Daily Operation Procedures

5.1 Power-On Sequence

1. Verify the battery pack is seated and the main breaker is OFF.
2. Flip the main breaker ON. The Pi5 boots; the RX72N status LED begins its breathing pattern within a second.
3. Wait approximately 30 seconds for ROS2, Caddy, and the cockpit API to come up (driven by `star-slam-mvp.service`).
4. Confirm Tailscale has assigned the Pi5 its tailnet address. (The Pi5 side has a 8s USB-enumeration delay before the SLAM MVP script starts.)
5. The robot is now in MANUAL mode with the emergency stop cleared. It will not move until you command it.

5.2 Opening the Cockpit

Open two browser tabs from the operator laptop:

1. The public Grafana URL (see Section 4). The STAR dashboard has control panels for autonomy, e-stop, speed, and SLAM, plus live telemetry gauges.
2. The public Lichtblick URL. The default layout loads the 3D scene, LiDAR, map, stereo point cloud, and IMU plots. In "Open Connection", enter `ws://100.64.0.6:8765`.

5.3 Granting the Browser Trust for `star.local`

The Grafana dashboard loads map images directly from `https://star.local/map.png`. This URL is served by the Pi5's Caddy using a local CA. Install that CA on the operator laptop one time (per `docs/ARCHITECTURE.md`), otherwise the map panel will show a broken-image placeholder.

5.4 Teleoperation (MANUAL mode)

In MANUAL mode the robot follows velocity commands from the Grafana cockpit or from a publisher on `/cmd_vel`. Two input options:

- **Grafana speed panel:** POST to `/api/speed` with a scalar in $[0, 1]$, then POST linear/angular velocity deltas to `/api/cmd_vel`.
- **USB gamepad** (at the workstation): left stick Y axis commands linear velocity (inverted), left stick X axis commands angular velocity; deadzone is 0.1; commands publish at 50 Hz.

The `star_safety_monitor` enforces a linear ceiling of 1.0 m/s and an angular ceiling of 2.0 rad/s. Commands outside this envelope are clamped.

5.5 Switching to AUTONOMY

From the Grafana cockpit, click the AUTONOMY toggle. This sends POST `/api/autonomy/toggle` to the cockpit API, which flips the latched `/star/autonomy_enable` topic. When AUTONOMY is on, the supervisor forwards Nav2's `/nav2/cmd_vel` to the gated `/cmd_vel_out` that `star_simple_bridge` consumes; operator teleop is ignored.

To give a Nav2 goal, click a target on the Lichtblick 3D panel configured to publish to `/goal_pose`. Nav2 plans and executes, publishing status back to the topic tree.

5.6 Running SLAM

If SLAM is not already active:

1. Click the SLAM START toggle in Grafana (POSTs to `/api/slam/start`). The cockpit API activates the `/slam_toolbox` lifecycle node.
2. Drive the robot slowly through the space. The Lichtblick map panel fills in as scans accumulate. A separate SLAM map can be viewed directly at `https://star.local/map.png`.
3. When finished, click SLAM STOP to deactivate the node. To reset the map, send POST `/map/reset?confirm=yes` to `https://star.local/map/reset` on `star_simple_bridge`.

5.7 Running a Compliance Audit

Compliance nodes (Section 7) publish check results on the `/compliance/*` topics whenever SLAM is active and the robot is moving. The ramp-slope check is [IMPLEMENTED]; other checks are [STRETCH] or [ARCHITECTED] per the SDD traceability matrix. The report generator in `final_capstone/compliance-engine/star_compliance/report/` produces a PDF audit report at the end of a run; copy it off the Pi5 over SCP.

5.8 Graceful Shutdown

1. From Grafana, clear AUTONOMY (back to MANUAL) and optionally SLAM STOP.
2. SSH to the Pi5 and run `sudo systemctl stop star-slam-mvp.service` to bring down ROS2 nodes gracefully.
3. Run `sudo shutdown -h now` on the Pi5.
4. Wait for the Pi5 power LED to extinguish, then flip the main breaker OFF.

6 Dashboard Reference

6.1 Grafana Cockpit

The committed dashboard is `monitoring/grafana-star-dashboard.json`. It contains row groups for SLAM map, motors, safety, sensors, and SLAM diagnostics. Key widgets the operator uses:

- **Map panel:** iframe-style HTML rendering of `https://star.local/map.png`. Auto-refreshes on the Grafana time-range tick.
- **Motor gauges:** per-wheel velocity in m/s (metric `star_motor_velocity_mps{wheel=...}`).
- **E-stop button and indicator:** POST `/api/estop/toggle`. The indicator gauge reflects the latched `/star/estop` topic.
- **Autonomy toggle:** POST `/api/autonomy/toggle`.
- **Speed slider:** POST `/api/speed` with a scalar in `[0,1]` that becomes `/star/speed_scale`.
- **SLAM start/stop buttons:** POST `/api/slam/start` or `/api/slam/stop`.

Grafana's minimum refresh interval is configured to 1 s via the `GF_DASHBOARDS_MIN_REFRESH_INTERVAL` environment variable on the Grafana pod.

6.2 Lichtblick Layout

The default layout `star-ros2/src/star_bringup/foxxglove/star-default.json` wires the following topics into Lichtblick's 3D panel:

- `/scan` – RPLiDAR C1 scan (visible, green, 5 px).
- `/map` – occupancy grid from `slam_toolbox`.
- `/cloud_map` – 3D point cloud, height-colored, 6 px.
- `/octomap_occupied_space` – Octomap (hidden by default; enable for obstacle review).
- `/stereo/points2` – stereo-camera point cloud, 4 px, rainbow-colored.
- `/tf` – transform tree.
- `/odometry/filtered` – EKF odometry, arrow marker.
- `/odom/unfiltered` – raw wheel odometry, arrow marker.

Plots beside the 3D panel show IMU acceleration and angular velocity on all three axes.

6.3 Foxglove Studio (desktop alternative)

Foxglove Studio can load the same layout JSON. Connect to `ws://100.64.0.6:8765` (the foxglove bridge on the Pi5). This is useful when the public Lichtblick URL is unavailable or for power users who prefer the desktop app.

7 Topic, Endpoint, and Port Reference

7.1 Ports Exposed on the Pi5

Table 2: Network ports exposed on the Pi5.

Port	Service	Role
443	Caddy (tls internal)	Fronts <code>star.local</code>
8765	foxglove_bridge	ROS2 over WebSocket
9100	node_exporter	Scraped by Alloy / Prometheus
9101	star_simple_bridge	<code>/map.*</code> , metrics, map reset
9102	star_cockpit_api	Cockpit HTTP API

7.2 Cockpit HTTP API

The following endpoints are exposed by `star_cockpit_api` on `100.64.0.6:9102` (and are reverse-proxied under `https://star.local/api/*` via Caddy).

Table 3: `star_cockpit_api` endpoints.

Method	Path	Purpose
GET	<code>/api/state</code>	Cached {autonomy, estop, speed, slam}
POST	<code>/api/autonomy/toggle</code>	Flip <code>/star/autonomy_enable</code>
POST	<code>/api/estop/toggle</code>	Flip <code>/star/estop</code>
POST	<code>/api/speed</code>	Body {"value":0.75} -> <code>/star/speed_scale</code>
POST	<code>/api/cmd_vel</code>	Body {"linear":0.5,"angular":0} -> <code>/cmd_vel</code>
POST	<code>/api/slam/start</code>	Configure + activate <code>/slam_toolbox</code>
POST	<code>/api/slam/stop</code>	Deactivate <code>/slam_toolbox</code>

7.3 ROS2 Topics the Operator Cares About

Table 4: Operator-relevant ROS2 topics.

Topic	Type	Notes
/cmd_vel	Twist	Manual teleop input
/nav2/cmd_vel	Twist	Nav2 output (AUTONOMY)
/cmd_vel_out	Twist	Gated output to bridge
/goal_pose	PoseStamped	Click target in Lichtblick
/star/autonomy_enable	Bool (latched)	MANUAL vs AUTONOMY
/star/estop	Bool (latched)	Soft e-stop
/star/speed_scale	Float32 [0, 1]	Speed governor
/odom/unfiltered	Odometry	Raw wheel odometry
/odometry/filtered	Odometry	EKF output
/imu/data	Imu	Fused BNO055 data
/scan	LaserScan	RPLiDAR C1
/map	OccupancyGrid	SLAM output
/compliance/report	Custom	Aggregate ADA result
/compliance/ramp_slope	Custom	Per-check result

8 Safety, Warnings, and Emergency Procedures

8.1 Software Emergency Stop

Three independent paths can stop the robot:

1. Click the E-STOP button on the Grafana dashboard (POSTs to `/api/estop/toggle`). This latches `/star/estop` to true and the supervisor gates `/cmd_vel_out` to zero.
2. Publish `true` directly on `/star/estop` (for example, from a command-line `ros2 topic pub` in an SSH session).
3. Physically disconnect USB to `/dev/star-mcu`; the `star_simple_bridge` stops delivering velocity commands and the RX72N watchdog zeroes the motor outputs within one control period.

8.2 Auto-E-Stop Thresholds

The `star_safety_monitor` ROS2 node enforces the limits in Table 5. Values come from `star-ros2/src/star_saf` and are authoritative.

Table 5: Safety-monitor thresholds.

Parameter	Value	Behavior
heartbeat_timeout_ms	500	Alarm if no heartbeat
max_linear_velocity	1.0	Clamped to 1 m/s
max_angular_velocity	2.0	Clamped to 2 rad/s
stall_detection_threshold	0.05	Below this counts as stall
stall_samples_required	5	Consecutive stalls -> E-stop
publish_rate	10.0	Diagnostics at 10 Hz
obstacle_estop_distance	0.10	Sonar < 10 cm -> E-stop
obstacle_warn_distance	0.30	Sonar < 30 cm -> warn
obstacle_clear_count_required	10	10 clear samples -> release
enable_auto_estop	true	Master enable
estop_recovery_delay	5.0	Delay before auto-recovery

8.3 Operator Safety Rules

- Never reach under the robot while the main breaker is ON.
- Keep the emergency-stop browser tab visible at all times during AUTONOMY.
- Do not lift the robot by the LiDAR dome or by any wire harness. Use the chassis mounts only.
- Before flashing firmware, set the robot on blocks so the wheels are off the ground.
- Never insert SCI-boot mode (SW4 Pin1 = ON) without a deliberate plan to reset the switches afterwards; the robot cannot run user firmware in that state.

9 Troubleshooting and Error Codes

9.1 Common Failure Modes

Problem: The Grafana cockpit map panel shows a broken image. The browser does not trust the Pi5 Caddy's local CA. Install the CA per docs/ARCHITECTURE.md and reload. If the problem persists, confirm `curl -k https://star.local/map.png` from any machine on the tailnet returns an image.

Problem: Grafana panels are empty. Prometheus is not receiving metrics. On the Pi5, run `systemctl status alloy` and confirm the Alloy agent is exporting to `http://100.64.0.1:30909/api/v1/write`. Restart with `sudo systemctl restart alloy`.

Problem: Lichtblick cannot connect to ws://100.64.0.6:8765. The foxglove bridge did not start. SSH to the Pi5 and check `systemctl status star-slam-mvp.service` and the log at `/tmp/slam-mvp-logs/`. Restart with `sudo systemctl restart star-slam-mvp.service`.

Problem: AUTONOMY toggle has no effect. The supervisor is not subscribed, or the cockpit API is down. `curl https://star.local/api/state` should return valid JSON. If not, restart with `sudo systemctl restart star-cockpit-api.service`.

Problem: Robot stalls and emergency stop engages. The safety monitor detected a wheel velocity below `stall_detection_threshold` for five consecutive samples. Check for mechanical binding, low battery, or a disconnected motor wire. After clearing the cause, wait `estop_recovery_delay` seconds and the monitor re-enables motion.

Problem: Firmware flash fails on E2 Lite. Confirm SW4 Pin1 = OFF and Pin2 = OFF. Confirm the board has external power (E2 Lite does not power the STAR PCB). Confirm FINE speed is 250 kbit/s; 1 Mbit/s is known to cause framing errors on our PCB.

9.2 RX72N Error Code Ranges

When a firmware error surfaces on a ROS2 diagnostic message, the `rx_err_t` code maps to a range:

- 0x000 – success.
- 0x100–0x1FF – generic error (null pointer, invalid argument, not initialized). Restart the bridge or power-cycle.
- 0x200–0x2FF – hardware error (SPI, I2C, GPIO). Check wiring and peripheral config.
- 0x300–0x3FF – RTOS error (ThreadX semaphore, mutex, queue). Reboot the RX72N.
- 0x400–0x4FF – communication error (CRC, framing, HARQ exhausted). Inspect SPI connector; reflash firmware if persistent.
- 0x500–0x5FF – validation error (out-of-range input, boundary check failed). Software-side bug; file an issue against the repo.

9.3 LED Status Patterns

The on-board LEDs managed by `star-rx72n-firmware/blinky/main.c` breathe sequentially across the six indicators as an "alive" heartbeat. A static (non-breathing) pattern indicates the RX72N firmware has halted; power-cycle the board and observe the boot LED sequence.

10 Maintenance

10.1 Firmware Update

1. On the workstation, `git pull` the repository.
2. Rebuild: `make build-rx72n-release`.
3. Set the robot on blocks so the wheels are free.
4. Flash via E2 Lite: `./scripts/flash-rx72n.sh star-rx72n-firmware/build/release/star-rx72n.mot`.
5. Observe the LED sequence and confirm heartbeat.
6. Power-cycle the board and verify the robot responds to a manual `cmd_vel` pulse in the Grafana dashboard.

10.2 Rebuilding the ROS2 Workspace

```
cd /workspaces/STAR      # or wherever the repo lives on the Pi5
./build-ros2.sh          # wraps colcon build with Cyclone DDS
sudo systemctl restart star-slam-mvp.service
```

10.3 Bring-Up Tests

Each hardware subsystem has a standalone smoke test under `star-rx72n-firmware/`:

- `motor_spin_test/` – spin each wheel open-loop.
- `pwm_test/` – scope the PWM waveform on the motor connectors.
- `encoder_test/` – print quadrature counts at runtime.
- `gpio_test/` – toggle every GPIO and verify with the external harness.
- `imu_test/` – print BNO055 orientation over UART.
- `uart_test/` – loopback test on the debug UART.
- `sonar_test/` – trigger and read HC-SR04 channels.

Each directory contains a CMake project; build and flash it in place of the main firmware, observe the expected behavior, then reflash the production image.

10.4 Updating the Dashboards

Grafana dashboard edits are stored back to `monitoring/grafana-star-dashboard.json`, committed, and pushed through the cluster admin’s Grafana API import flow. Lichtblick layouts are distributed through the k3s ConfigMap-injected default layout; to edit, change `star-ros2/src/star_bringup/foxglove` commit, rebuild the Lichtblick deployment image, and re-apply.

10.5 Updating Firmware PID Gains

PID gain design is performed offline in MATLAB per the SDD. New gains are written to `star-rx72n-firmware/...` picked up by the firmware build, and deployed via the firmware-update procedure. There is no on-line autotuning path.

11 Assumptions and Limitations

Operator-facing subset of the SDD limitations:

- Four motors maximum; firmware is sized to exactly four GPTW and four TPU channels.
- Single operator at a time; the Grafana cockpit has no role separation or contention arbitration.
- Tailnet required; the cockpit is not reachable outside the Tailscale mesh and the public Cloudflare entry point.
- English only; no internationalization.
- Online PID autotuning is not supported. Plant or load changes require an offline MATLAB re-run and a firmware rebuild.
- Compliance coverage: only the ramp-slope check is [IMPLEMENTED]; path width and trip hazard are [STRETCH], and ramp width, ramp landings, door clear width, and door threshold are [ARCHITECTED] per the SDD traceability matrix.
- `star-simple` (a planned lightweight bring-up tree on the field Pi5) is [ARCHITECTED]: some of it is present on the physical robot but has not been committed back to the repository yet. Use `star_simple_bridge` directly for now.

- Grafana dashboards are deployed by the cluster admin through a Grafana HTTP API import; the operator cannot rebuild the dashboard without cluster access.

12 Glossary

ADA 405.2

Section 405.2 of the 2010 ADA Standards, which limits ramp slope to 1:12 rise-over-run.

Caddy HTTP reverse proxy used in two places in STAR: on the Pi5 with `tls internal` (local CA) to front `star_simple_bridge` and `star_cockpit_api`; and on a separate homelab LXC as the public entry point behind Cloudflare.

Foxglove Studio

Desktop ROS2 visualization application. STAR ships a Foxglove-compatible JSON layout.

Lichtblick

Open-source web fork of Foxglove Studio. STAR's primary operator-facing 3D visualization, deployed on k3s.

Grafana

Browser-based telemetry dashboard. STAR's primary cockpit.

Tailscale

WireGuard-based VPN mesh. STAR's Pi5 lives at `100.64.0.6`; the homelab cluster at `100.64.0.1`.

k3s Lightweight Kubernetes distribution used to host Grafana, Prometheus, and Lichtblick in STAR's homelab cluster.

Alloy Grafana Labs telemetry agent installed on the Pi5 that remote-writes metrics to Prometheus.

SLAM MVP

Minimal SLAM stack launched by `star-slam-mvp.service` at boot. Uses `slam_toolbox` in async mode.

Supervisor

ROS2 node that gates `/cmd_vel` and `/nav2/cmd_vel` into `/cmd_vel_out` based on latched autonomy and e-stop topics.

E2 Lite

Renesas JTAG / FINE debug probe used to program the RX72N.

Appendices

Appendix A: Quick Reference Card

Power on: Flip breaker -> wait 30 seconds for services -> open Grafana and Lichtblick tabs.

Emergency stop: Click E-STOP in Grafana, or `curl -X POST https://star.local/api/estop/toggle`.

MANUAL mode drive: Grafana speed slider + `cmd_vel` POST, or USB gamepad at the workstation.

AUTONOMY mode: Click AUTONOMY toggle in Grafana, then click a goal in Lichtblick.

SLAM start: Click SLAM START in Grafana; drive slowly; watch `/map.png` fill in.

Flash firmware: `./scripts/flash-rx72n.sh star-rx72n.mot` (E2 Lite default).

Shutdown: Clear AUTONOMY -> `stop star-slam-mvp.service` -> `sudo shutdown -h now` -> flip breaker OFF.

Appendix B: Config File Cheat Sheet

File	Purpose
<code>star-ros2/src/star_bringup/config/slam_toolbox.yaml</code>	SLAM tuning (resolution, loop closure)
<code>star-ros2/src/star_bringup/config/nav2_params.yaml</code>	Nav2 planner + controller parameters
<code>star-ros2/src/star_bringup/config/ekf.yaml</code>	robot_localization EKF parameters
<code>star-ros2/src/star_bringup/config/explore_params.yaml</code>	<code>explore_lite</code> frontier search parameters
<code>star-ros2/src/star_safety_monitor/config/safety_monitor.yaml</code>	Safety-monitor thresholds (Table 5)
<code>infra/pi5/etc/caddy/Caddyfile</code>	Pi5-side Caddy (<code>star.local</code> , <code>tls internal</code>)
<code>monitoring/grafana-star-dashboard.json</code>	Grafana dashboard definition
<code>star-ros2/src/star_bringup/foxglove/star-default.json</code>	Lichtblick / Foxglove layout

Appendix C: Important Addresses and Credentials

- Pi5 tailnet address: `100.64.0.6` (hostname `star.local`).
- Homelab cluster tailnet address: `100.64.0.1` (Prometheus remote-write `http://100.64.0.1:30909/api/v1/w`).
- Pi5 serial device for RX72N: `/dev/star-mcu` (udev-aliased to `/dev/ttyACM0`).
- Pi5 serial device for RPLiDAR C1: `/dev/star-lidar`.
- Local-CA trust: install the Pi5 Caddy local CA on every operator laptop (one-time, per `docs/ARCHITECTURE.md`).
- Public URLs for Grafana and Lichtblick are deployed via Cloudflare and the homelab public Caddy; consult the cluster admin for the current host names.

Appendix D: Change Log

- Version 1.0, April 28, 2026 – Initial release for ESET 420 capstone submission.